PROJECT: WEBSITE AND APP ANALYTICS USING POWER BI

INTRODUCTION: This project focuses on analysing website and app performance using Power BI to gain meaningful insights from user data. It includes collecting, cleaning, and visualizing data such as website traffic, app usage, downloads, sessions, engagement, and user behaviour. Interactive dashboards, charts, filters, and reports are used to identify trends, monitor performance, improve user experience, and support effective data-driven decision-making.

Objective 1: Understand the Website and App Data Structure and Key Performance Metrics

- Review datasets and data sources.
- Identify important fields and Key Performance Indicators (KPIs).
- Understand user, session, traffic, and engagement-related data.
- Explore data categories, attributes, and business requirements.

Features / Functions Used: Data View| Data Profiling| Summary Statistics| Filters

Objective 2: Pre-process and Clean the Data for Accurate Analysis

- Check for missing values.
- Remove duplicate records.
- Handle inconsistencies in data.
- Correct data formats and data types.
- Transform and standardize data where required.

Features / Functions Used: Power Query Editor |Replace Values |Remove Duplicates| Data Type Conversion |Conditional Columns

Objective 3: Create Data Models and Establish Relationships Between Tables

- Connect related tables.
- Define primary and foreign key relationships.
- Create a structured data model.
- Validate relationships for accurate analysis.

Features / Functions Used: Relationship View| Data Modelling| Manage Relationships| Star Schema Concepts

Objective 4: Create Calculated Fields and DAX Measures to Support Analysis

- Create custom columns.
- Create calculated columns.
- Develop key performance metrics and KPIs.
- Calculate session duration, bounce rate, conversion rate, and other business metrics.

Features / Functions Used: Custom Columns| Calculated Columns| DAX Functions| COUNT()| DISTINCTCOUNT()| SUM()| DIVIDE()|IF()

Objective 5: Analyse User Behaviour and Website/App Performance

- Track page views and user actions.
- Analyse user navigation paths and engagement.

- Compare new and returning users.
- Identify major traffic sources.
- Measure conversions, exits, and drop-off points.

Functions Used: Group By| Segmentation| DAX Measures| Aggregation Functions| Data Analysis Techniques

Objective 6: Create Visualizations to Identify Trends and Patterns

- Visualize user behaviour and engagement.
- Compare different user segments.
- Track website and app performance over time.
- Analyse conversions and traffic sources.
- Present key insights through visual representations.

Functions Used: Bar Charts| Line Charts| Funnel Charts| Pie/Donut Charts| KPI Cards| Tables and Matrix Visuals| Slicers and Filters

Objective 7: Develop Interactive Reports and Dashboards for Decision-Making

- Design dashboard pages.
- Add interactive visuals and filters.
- Implement drill-through functionality.
- Create multi-page reports.
- Prepare the final project presentation and dashboard.

Features / Functions Used: Power BI Dashboard| KPI Cards| Slicers| Drill Through| Bookmarks| Multi-page Reports| Storytelling Visuals